TATA CARA PENGOPERASIAN SISTEM TRANSMISI & OVERLAY STUDIO

Pedoman teknis komprehensif tahapan produksi siaran langsung: Ingest Kontribusi Lapangan, Penataan Branding Station ID & Lower-Third, Pengaturan Routing Restreaming Multi-Tujuan (Fan-out), dan Penarikan Feed di Studio Master (MCR).

1. PARAMETER KUNCI & KREDENSIAL AKSES SISTEM

Akses Web Dashboard & Monitoring:
- **URL Dashboard:** http://103.89.5.211:8080
- **Autentikasi:** Username: admin | Password: [REDACTED]
- **Mobile Controller:** Unduh installer Android di /download/apk

Alokasi Port Penyiaran & Feed Preview Engine:
- **SRT Gateway Port:** UDP 8890 (ARQ Latency Buffer: 2000 ms)
- **RTMP Ingest Port:** TCP 1935 | **MediaMTX API:** Port 9997
- **WebRTC Low-Latency (<300ms):** http://103.89.5.211:8889/live/[KEY]

2. TAHAP 1 — INGEST & KONTRIBUSI SIARAN (PENGIRIM / KAMERA / ENCODER)

Opsi A: Kamera Smartphone (Makasna Remote App)
1. Buka aplikasi **Makasna Remote** di HP Android (Camera2 API Native).
2. Hubungkan ke host: 103.89.5.211:8080 | User: admin | Pass: [REDACTED].
3. Masuk ke tab **Camera**, tentukan **Stream Key** (cth: cam1 atau lapangan).
4. Atur eksposur/ISO, framing 16:9, lalu tekan tombol **START STREAM**.

Opsi B: Hardware Encoder / vMix / OBS Lapangan

Kirimkan stream audio-video siaran dari perangkat lapangan:
- **Protokol SRT Caller (Direkomendasikan - Tahan Jitter & Drop):**

```
srt://103.89.5.211:8890?streamid=publish:live/[STREAM_KEY]&latency=2000000
```

- **Protokol RTMP Ingest (Alternatif):** rtmp://103.89.5.211:1935/live (Key: [STREAM_KEY])

Ganti [STREAM_KEY] dengan ID feed sumber (Contoh: cam1, vMix_lapangan).

Parameter Video	H.264 / AVC (High Profile 4.1), 1080p / 720p @ 30/60 fps	**Bitrate & Rate Control**	3.000 - 6.000 kbps (Wajib CBR, Keyframe 2s)
Parameter Audio	AAC-LC Stereo, 128 - 192 kbps, Sample Rate 48 kHz	**Standar Level Audio**	Target Level -18 dBFS (0 VU), Ceiling Clip 0 dBFS

3. TAHAP 2 — BRANDING SIARAN (STATION ID & OVERLAY STUDIO)

Alur Penataan Grafis & Watermark Siaran:
1. Masuk ke Web Dashboard dan pilih tab menu **Station ID / Overlays**.
2. **Workspace Kanvas Monitor 16:9:** Geser posisi grafis siaran secara visual (drag-and-drop mouse/sentuhan) atau melalui slider presisi koordinat dua arah:
- **Layer 1 (L1 - Station ID Logo):** Atur posisi pojok atas (Preset: *Top-Right* atau *Top-Left*). Rekomendasi Skala: 14-18%, Opacity: 95%.
- **Layer 2 (L2 - Lower-Third / Live Bug):** Atur posisi banner bawah (Preset: *Lower-Third News*). Skala: 18-25%, Opacity: 100%.
3. **Aktivasi Engine Restream Egress:** Aktifkan saklar switch **'Burn-in on YouTube / RTMP Egress'** agar FFmpeg merender grafis langsung ke bingkai video secara *real-time* via libx264 zerolatency filter.
4. **Penyimpanan Parameter:** Klik tombol **Save Configuration**. Koordinat otomatis tersimpan permanen di server (overlay_config.json).

4. TAHAP 3 — MANAJEMEN ROUTING & RESTREAMING FAN-OUT (DISTRIBUSI)

Prosedur Distribusi Aliran Siaran (Routes Workspace):
1. Masuk ke tab **Routes Workspace** pada Web Dashboard.
2. Klik tombol **+ Add Route** (atau pilih tombol *Edit* pada rute yang telah tersedia):
- **Source Ingest:** Tentukan nama rute (cth: Main-Broadcast-Route) dan path feed sumber (cth: live/cam1).
- **Destinasi 1 (Restream YouTube Live RTMP):** Masukkan target URL rtmp://a.rtmp.youtube.com/live2/[REDACTED].
- **Destinasi 2 (SRT Egress ke Master Control / Mitra):** Masukkan target srt://[IP_TUJUAN]:8890?streamid=publish:live/[STREAM_KEY]&latency=2000000.
3. **Mulai Transmisi:** Tekan tombol **Start Route**. Indikator berubah menjadi hijau (Active).
4. **Pemantauan Telemetri:** Pantau kestabilan *Egress Bitrate*, *FPS*, *Total Volume Terkirim*, dan *ARQ Resend Rate* pada kartu rute.

5. TAHAP 4 — PENERIMAAN FEED DI STUDIO MASTER (RECEIVER / MCR / vMix / OBS)

A. Penarikan Sinyal di vMix (Studio Master Control)
1. Buka vMix → klik tombol **Add Input** di pojok kiri bawah.
2. Pilih menu **Stream / SRT**, lalu aktifkan tab **SRT (Caller)**.
3. Masukkan parameter koneksi:
- **Hostname / IP:** 103.89.5.211 | **Port:** 8890
- **Stream ID:** read:live/[STREAM_KEY] *(Contoh: read:live/cam1)*
- **Latency:** 2000 ms (Buffer pemulihan paket 2 detik)
4. Klik **OK**. Sinyal siaran audio/video langsung aktif pada vMix.

B. Penarikan Sinyal di OBS Studio
1. Pada panel **Sources**, klik ikon **+** → pilih **Media Source**.
2. Beri nama input, lalu **hilangkan centang** pada opsi *'Local File'*.
3. Pada kolom **Input**, ketik URL penarikan SRT caller:

```
srt://103.89.5.211:8890?streamid=read:live/[STREAM_KEY]&latency=2000000
```

4. Pada kolom **Input Format**, ketik mpegts (atau kosongkan).
5. Klik **OK**. Siaran langsung otomatis tampil di canvas pratinjau.

CHECKLIST PRA-SIARAN, MONITORING & MITIGASI GANGGUAN (TROUBLESHOOTING):
- **Checklist Pra-Siaran (H-30 Menit):** Pastikan bandwidth upload pengirim memiliki margin ≥30% di atas bitrate; uji kesesuaian audio AAC-LC 48 kHz; pastikan Stream Key pengirim (publish:live/[KEY]) sinkron dengan sisi penarikan (read:live/[KEY]).
- **Monitoring On-Air:** Pantau Web Dashboard (port 8890). Pastikan status Ingest In bertambah, Packet Loss 0%, RTT stabil, dan bitstream stabil.
- **Mitigasi Video Putus/Patah:** Periksa stabilitas koneksi upload pengirim dan pastikan buffer ARQ disetel 2.000 ms. Bila overlay tidak terpasang di YouTube, pastikan toggle *Burn-in* aktif lalu klik *Save Configuration* dan restart rute.
- **Pasca-Siaran:** Tekan tombol **Stop Route** pada dashboard dan matikan transmisi encoder lapangan untuk menghemat resource server.

TRANSMISSION SYSTEM & OVERLAY STUDIO OPERATING PROCEDURE

Comprehensive technical operating guidelines for live broadcast pipelines: Field Contribution Ingestion, Station ID & Lower-Third Graphics Studio, Multi-Destination Restreaming Routing (Fan-out), and MCR Master Studio Feed Reception.

1. KEY PARAMETERS & SYSTEM ACCESS CREDENTIALS

Web Dashboard & Telemetry Monitoring:

- **Dashboard URL:** http://103.89.5.211:8080
- **Authentication:** Username: admin | Password: [REDACTED]
- **Mobile Controller:** Direct APK download via dashboard at /download/apk

Broadcast Ports & Low-Latency Preview Engines:

- **SRT Gateway Port:** UDP 8890 (ARQ Latency Buffer: 2000 ms)
- **RTMP Ingest Port:** TCP 1935 | **MediaMTX API:** Port 9997
- **WebRTC Ultra-Low Latency (<300ms):** http://103.89.5.211:8889/live/[KEY]

2. STAGE 1 — BROADCAST INGEST & CONTRIBUTION (SENDER / CAMERA / ENCODER)

Option A: Smartphone Camera (Makasna Remote App)

1. Launch **Makasna Remote** on Android (Native Camera2 API Engine).
2. Connect to gateway: 103.89.5.211:8080 | User: admin | Pass: [REDACTED].
3. Navigate to **Camera** tab, configure **Stream Key** (e.g., cam1 or field).
4. Adjust exposure/ISO, lock 16:9 framing, then tap **START STREAM**.

Option B: Hardware Broadcast Encoder / vMix / OBS

Publish live contribution stream from OB-van or field encoder:

- **SRT Caller Protocol (Recommended - High Packet Loss Immunity):**

```
srt://103.89.5.211:8890?streamid=publish:live/[STREAM_KEY]&latency=2000000
```

- **RTMP Ingest Protocol (Fallback):** rtmp://103.89.5.211:1935/live (Key: [STREAM_KEY])

Replace [STREAM_KEY] with feed source identifier (e.g., cam1, vMix_field).

Video Specification	H.264 / AVC (High Profile 4.1), 1080p / 720p @ 30/60 fps	**Bitrate & Rate Control**	3,000 - 6,000 kbps (Strict CBR, 2s Keyframe)
Audio Specification	AAC-LC Stereo, 128 - 192 kbps, Sample Rate 48 kHz	**Standard Audio Level**	Target Level -18 dBFS (0 VU), Ceiling Peak 0 dBFS

3. STAGE 2 — BROADCAST BRANDING (STATION ID & OVERLAY STUDIO)

On-Air Graphics Composition & Watermarking Workflow:

1. Log in to the Web Dashboard and access the **Station ID / Overlays** workspace tab.
2. **Interactive 16:9 Confidence Monitor Canvas:** Position graphic layers visually (drag-and-drop mouse/touch) or via precision coordinate sliders:
- **Layer 1 (L1 - Station ID Logo):** Position at top corner (Presets: *Top-Right* or *Top-Left*). Recommended Scale: 14-18%, Opacity: 95%.
- **Layer 2 (L2 - Lower-Third / Live Bug):** Position at lower banner area (Preset: *Lower-Third News*). Scale: 18-25%, Opacity: 100%.
3. **Master Restream Egress Activation:** Turn on the **'Burn-in on YouTube / RTMP Egress'** switch so FFmpeg burns graphics directly into live video frames via libx264 zerolatency filter.
4. **Save Parameters:** Click **Save Configuration**. Graphic coordinates are permanently stored on the gateway server (overlay_config.json).

4. STAGE 3 — ROUTING MANAGEMENT & FAN-OUT RESTREAMING (DISTRIBUTION)

Stream Distribution & Fan-out Execution (Routes Workspace):

1. Open the **Routes Workspace** tab within the Web Dashboard.
2. Click **+ Add Route** (or click *Edit* on an existing routing profile):
- **Source Ingest Binding:** Define route identifier (e.g., Main-Broadcast-Route) and source feed path (e.g., live/cam1).
- **Destination 1 (YouTube Live RTMP Restream):** Input target URL rtmp://a.rtmp.youtube.com/live2/[REDACTED].
- **Destination 2 (SRT Egress to Master Control / Affiliate):** Input target srt://[TARGET_IP]:8890?streamid=publish:live/[STREAM_KEY]&latency=2000000.
3. **Start Transmission:** Click **Start Route**. The route status indicator illuminates green (Active).
4. **Telemetry Monitoring:** Supervise live stability of *Egress Bitrate*, *FPS*, *Total Volume Sent*, and *ARQ Resend Rate*.

5. STAGE 4 — MASTER STUDIO FEED RECEPTION (RECEIVER / MCR / vMix / OBS)

A. Pulling Feed in vMix (Studio Master Control)

1. Launch vMix → click **Add Input** located in bottom-left corner.
2. Select **Stream / SRT** menu, then open **SRT (Caller)** tab.
3. Configure connection parameters:
- **Hostname / IP:** 103.89.5.211 | **Port:** 8890
- **Stream ID:** read:live/[STREAM_KEY] *(e.g., read:live/cam1)*
- **Latency:** 2000 ms (2-second ARQ recovery buffer)
4. Click **OK**. Live audio/video stream locks instantly onto vMix.

B. Pulling Feed in OBS Studio

1. In **Sources** panel, click **+** icon → select **Media Source**.
2. Label the source, then **uncheck** *'Local File'* option.
3. In **Input** field, enter full SRT caller URL syntax:

```
srt://103.89.5.211:8890?streamid=read:live/[STREAM_KEY]&latency=2000000
```

4. Set **Input Format** to mpegts (or leave empty).
5. Click **OK**. Live feed renders onto OBS preview canvas.

PRE-BROADCAST CHECKLIST, MONITORING & RAPID MITIGATION (TROUBLESHOOTING):

- **Pre-Broadcast Checklist (T-30 Minutes):** Verify uplink bandwidth has ≥30% headroom above stream bitrate; test AAC-LC 48 kHz compliance; verify identical Stream Keys between sender (publish:live/[KEY]) and receiver (read:live/[KEY]).
- **Live On-Air Supervision:** Monitor Web Dashboard (port 8890). Confirm Ingest In counter increments, Packet Loss maintains 0%, and RTT stays within normal margins.
- **Video Freeze Mitigation:** Verify field upload stability and ensure ARQ buffer is set to 2,000 ms. If overlay does not appear on YouTube restream, verify *Burn-in* toggle is ON, click *Save Configuration*, and restart route.
- **Post-Broadcast Protocol:** Click **Stop Route** on dashboard and terminate field encoder transmission to release hardware compute resources.